

A Fixed-Source Cardinality-Monotonicity Obstruction for Native Prime-Shell Budgets

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Abstract

The source-first growing-grid audit of TPC-302 found a stable gap between a signed weighted target and an all-positive control, but did not determine how the weighted native budget itself changes with shell size. We test the stronger cardinality-only monotonicity shortcut on a fixed source spine: $(N, H, z) = (512, 58, 5)$ and $Q = 50, 60, 70, 90$, with shell cardinalities 10, 13, 15, 17. We use the three TPC-302 relative RMS tolerances, two kernel exponents, and three source normalizations. Outward interval comparisons give 21 strict descents and 33 strict ascents among 54 adjacent transitions, with no unresolved case. All 18 parameter series are nonmonotone, and nine descents occur at an unchanged common profile prefix. The strongest same-prefix contraction is below 0.284422. This is a scoped finite obstruction, not a refutation of an eventual asymptotic lower bound.

1 Question

TPC-302 transported a native weighted/positive budget gap to a 34-row growing/control grid and regenerated its signed target from the physical Gram matrix [1]. A tempting next shortcut would be to infer that the weighted native budget must grow whenever shell cardinality grows. Such an inference is stronger than the TPC-302 gap statement and needs its own test.

We freeze source scale $N = 512$, height $H = 58$, comparison cutoff $z = 5$, and compare the moving shells indexed by $Q = 50, 60, 70, 90$. Their cardinalities are 10, 13, 15, 17. The shells are not nested; the claim under attack is therefore cardinality-only monotonicity, not monotonicity under set inclusion.

For a common profile prefix k and target b , TPC-302 uses

$$B_{k,\tau}(b) = \min\{c^T M_k c : \|V_k c - b\|_2 \leq \tau \|b\|_2\}.$$

We read the common-prefix weighted budget, normalized by $\|\beta\|_2^2$, $\text{tr}(M_k)/k$, or $M_k[1, 1]$.

2 Interval certificate

Proposition 1 (Certified order). *If $B_j \in [L_j, U_j]$ and $B_{j+1} \in [L_{j+1}, U_{j+1}]$, then $U_{j+1} < L_j$ implies $B_{j+1} < B_j$, while $L_{j+1} > U_j$ implies $B_{j+1} > B_j$.*

Proof. Every represented value lies between its interval endpoints. The strict endpoint inequalities place the two values in disjoint ordered intervals. $\square$

Proposition 2 (Finite scoped refutation). *One certified descent refutes nondecreasing cardinality-only budget on the declared finite Q -spine.*

Table 1: Adjacent transition census on the fixed-source spine.

quantity	count	total	fraction
certified descents	21	54	21/54
certified ascents	33	54	33/54
unresolved overlaps	0	54	0/54
nonmonotone parameter series	18	18	18/18
same-prefix descents	9	54	9/54

Proof. The definition of nondecreasing requires every adjacent pair to have the later value at least the earlier value. A certified strict descent violates that condition. $\square$

A descent with equal common prefix k is especially informative: it cannot be blamed on adding or removing profile directions. It can still reflect the moving physical shell or the source-first change in the signed target.

3 Results

The strongest descent overall is the $Q = 60 \rightarrow 70$ transition at exponent 2, $\tau = 1/2$, normalized by $\|\beta\|_2^2$; its right/left interval ratio is below 0.224974. The strongest same-prefix descent is also $Q = 60 \rightarrow 70$ at exponent 2 and $\tau = 3/4$, with both common prefixes equal to $k = 6$ and ratio below 0.284422. The three normalizers give the same qualitative result, and each exponent–tolerance–normalizer series contains both an ascent and a descent.

These are interval-certified finite comparisons inherited from a frozen source-first certificate. They do not say that a suitably normalized budget cannot eventually obey a lower bound, nor do they turn moving-shell data into a nested-shell theorem.

4 Obstruction and next question

The result closes a useful but invalid shortcut: shell cardinality alone does not control the direction or size of the native weighted budget on this declared spine. The same-prefix cases further show that profile dimension is not the only explanation. The next natural experiment is to transport the weighted sign labels across overlapping shells, separating target-label switching from the physical operator/shell change. Uniform profile-budget growth, arithmetic L^2 , fixed-power credit, full Gate B, and a twin-prime conclusion remain open.

Reproducibility and scope

The canonical TPC-303 certificate stores every interval and transition. An independent checker reconstructs the census from TPC-302 without importing the producer, and a stress suite checks the interval logic. The Session-named Route-A/Route-B evaluator files are absent from this checkout; no official evaluator pass is asserted.

References

- [1] Liang Wang. Source-first growing-shell stability of the native weighted/positive budget gap, 2026. TPC-302 project release in the accompanying repository.